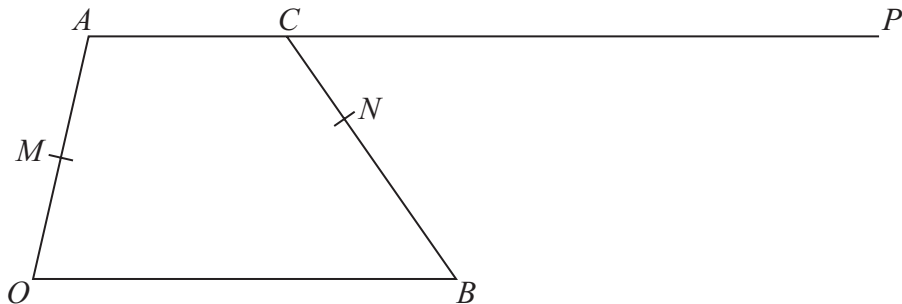


- 23** $OACB$ is a quadrilateral.
 ACP is a straight line.



M is the midpoint of OA .

N is the point on BC such that $BN:NC = 5:3$

$$\overrightarrow{OA} = \mathbf{a} \quad \overrightarrow{OB} = 3\mathbf{b} \quad \overrightarrow{AC} = 2\mathbf{b}$$

$$\overrightarrow{CP} = k \times \overrightarrow{AC} \quad \text{where } k \text{ is a scalar.}$$

Given that MNP is a straight line, find the value of k .
 You must show all your working.

$$k =$$

(Total for Question 23 is 5 marks)